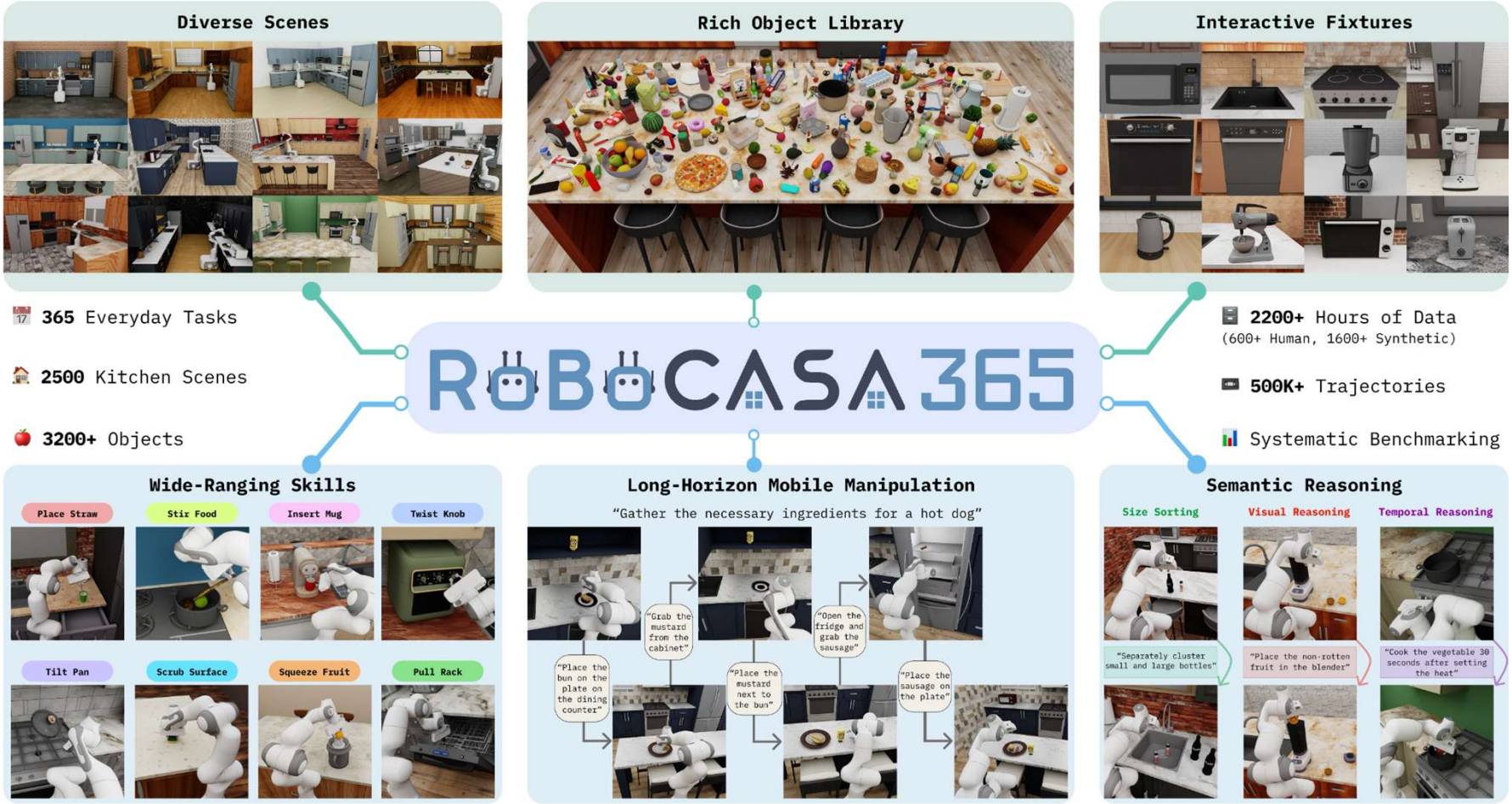
What is robocasa?

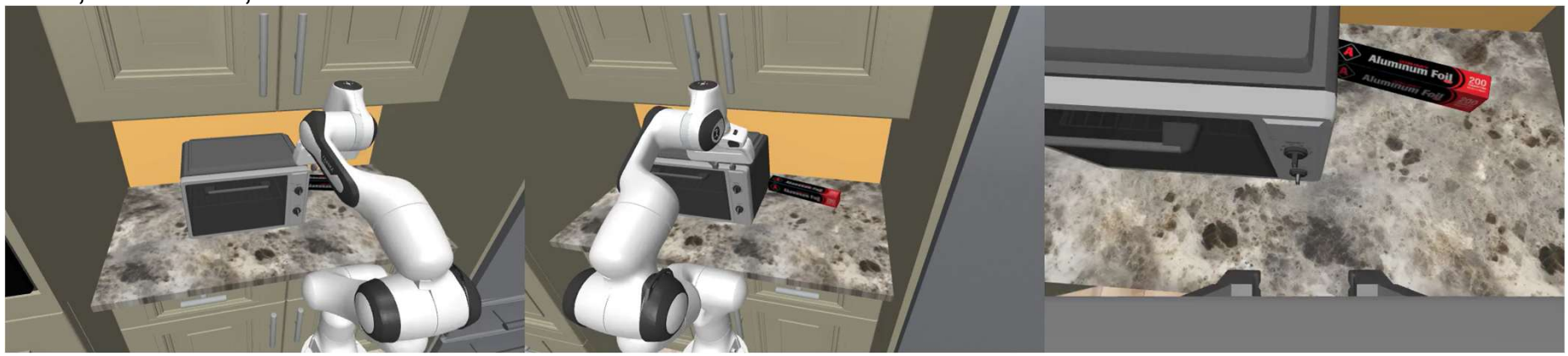
https://robocasa.ai/docs/build/html/introduction/overview.html



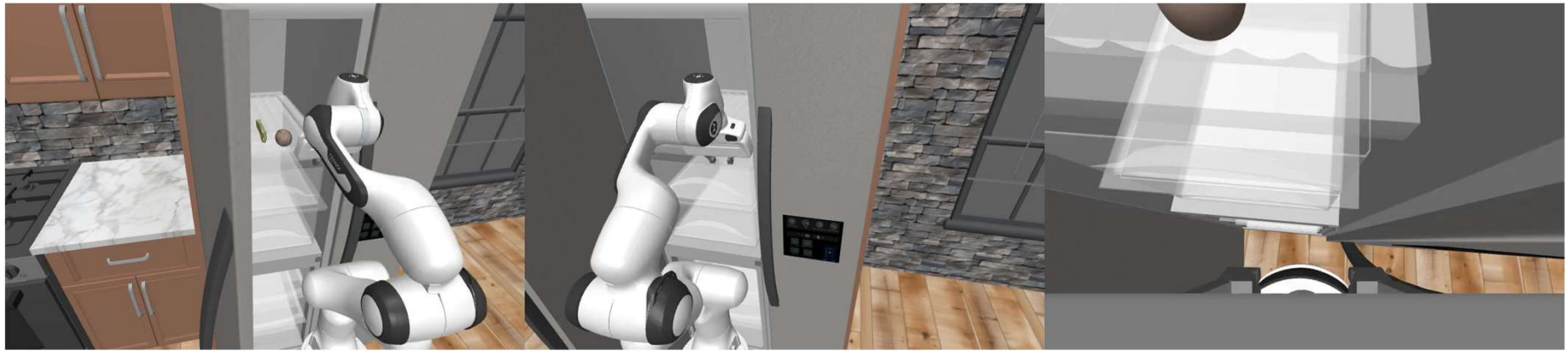
Tasks in Robocasa?

https://robocasa.ai/docs/build/html/tasks/composite_tasks.html

Atomic task: close door, open door, close lid, open lid, insertion, pick&place, press button, slide rack, turn lever, turn knob.



Composite tasks: The tasks that composed with atomic task.



Objective

- Using the robot PandaOmron to accomplish an “atomic task” in robocasa.(exclude the task navigation)
- Suggested Methods：
  - Reward shaping
  - Curriculum learning
  - Combine imitation learning and reinforcement learning

Installation https://robocasa.ai/

1. Set up conda environment:

```
conda create -c conda-forge -n robocasa python=3.11
```

2. Activate conda environment:

```
conda activate robocasa
```

3. Clone and setup robosuite dependency:

> **Important!**
> The latest version of RoboCasa uses the master branch of robosuite

```
git clone https://github.com/ARISE-Initiative/robosuite
cd robosuite
pip install -e .
```

4. Clone and setup this repo:

```
cd ..
git clone https://github.com/robocasa/robocasa
cd robocasa
pip install -e .
```

> **Installation troubleshooting**
>
> If you encounter issues with installing the numpy or nu
>
> `-c numba numba=0.56.4 -y`

5. Install the package and download assets:

```
python -m robocasa.scripts.setup_macros
python -m robocasa.scripts.download_kitchen_assets
```

Submission

- Code:
  - a zip file
  - a readme inside to describe how to evaluate the model and reduplicate the results.
  - A trained model
- Report:
  - Which method is used? The details?
  - What difficulty did you meet? How to solve?
  - The training curve of the robot
  - Visualization of the task